CURRICULUM VITAE

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ADDRESS

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EDUCATION

09/2026 – 07/2029	Doctor of Philosophy in Civil Engineering, Department of Civil Engineering, The University of Hong Kong (HKU), Hong Kong (expected)
09/2024 – 07/2026	Master of Philosophy in Civil Engineering, Department of Civil Engineering, The University of Hong Kong (HKU), Hong Kong (expected)
09/2022 – 09/2023	Master of Science in Computer Science, The University of Hong Kong (HKU), Hong Kong; GPA: 3.47
09/2017 – 06/2021	Bachelor of Engineering in Automation, Beijing Institute of Technology (北京理工大学), Beijing, China; GPA: 3.60

PROFESSIONAL EXPERIENCE

11/2023 – 08/2024	Research Assistant, The Chinese University of Hong Kong (CUHK), Hong Kong; full-time
09/2022 – 08/2024	Research Assistant, The University of Hong Kong (HKU), Hong Kong; part-time

AREAS OF RESEARCH

· Reinforcement learning and deep learning for multimodal transportation
· Application of Large language models in smart transportation

PUBLICATIONS

A. Journal Papers

(1) Chen, T., **Shen, Z.**, Feng, S., Yang, L., and Ke, J., 2025. Dynamic adjustment of matching radii under the broadcasting mode: a novel multi-task learning strategy and temporal modeling approach.
Transportation Research Part E: Logistics and Transportation Review, 193, 103822.

(2) Chen, T., **Shen, Z.**, Zhou, B., Liu, Y., Wang, S., and Ke, J., 2026. Addressing the online incremental transport mode choice prediction problem with an LLM-augmented class-incremental learning approach. *Transportation Research Part C: Emerging Technologies, 188, 105709*

B. Conference Papers / Proceedings

(1) **Shen, Z.**, Mu, Z., and Li, X., 2023. Multi-strategy collaborative optimized YOLOv5s and its application in distance estimation. AEECA 2023.

C. Working Papers

(1) **Shen, Z.**, Zhou, B., Chen, T., Wang, J., and Ke, J., 2026. ReLMM-TG: Few-Shot Tabular Generation via Reinforcement Learning over LLM-Compiled Mechanism Memory. *Artificial Intelligence for Transportation (under review)*

(2) **Shen, Z.**, Chen, T., Zhou, B., Wang, J., and Ke, J., 2026. A Deep Reinforcement Learning Model for Centralized Route Recommendation in Multi-modal Transit Networks. *Transportation Research Part E: Logistics and Transportation Review (Minor).*

(3) Wang, J., Chen, T., **Shen, Z.**, Liang, J., Zhou, B. and Ke, J. SmartSim: A Scalable and Multimodal Open-Source Mesoscopic Urban Traffic Simulator *Frontiers of Engineering Management (Major).*

D. Selected Conference Presentations

(1) **Shen, Z.**, Chen, T., Zhou, B., Wang, J., and Ke, J., 2026. A temporally aware deep reinforcement learning framework for centralized multi-path recommendation in large-scale multimodal transit networks. Presented at the 105th Transportation Research Board (TRB) Annual Meeting, Washington, DC, United States, January 11–15, 2026.

(2) **Shen, Z.**, Chen, T., Zhou, B., Wang, J., and Ke, J., 2025. Multipath: Deep learning based multimodal route guidance with user preference integration. Proceedings of the 29th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2025): Smart Mobility, Hong Kong, December 8–9, 2025.

(3) **Shen, Z.**, Chen, T., Wang, J., and Ke, J., 2024. Personalized fair matching in peer-to-peer ridesharing platforms under broadcasting mode: a llm-driven driver approach. Proceedings of the 28th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2024): Smart Mobility, Hong Kong, December 9–10, 2024.

RESEARCH PROJECTS

(1) Smart Traffic Fund (STF) of Hong Kong SAR Government, PSRI/78/2311/RA, "*SmartSim: AI-assisted Simulation Software for Multimodal Transportation Operations*", 1 Sep 2024 –31 August 2026, Core Member.

(2) Smart Traffic Fund (STF) of Hong Kong SAR Government, PSRI/29/2201/PR, "*Development of a Simulation Platform and Artificial Intelligent Algorithms for Optimising Operation and Management of Taxi E-hailing Services*", 31 Mar 2023 – 30 Sep 2024, Core Member.

(3) Environment Conservation Fund (ECF) Environmental Research, Technology Demonstration and Conference Projects, 102/2022, "*Estimating carbon emissions, assessing decarbonization strategies and managing green transportation in Hong Kong with a multifunctional simulation platform*", 1 March 2024 – 28 Feb 2026, Core Member.

(4) MTR Research Funding (MRF) 2025, HKU-25003, "*Multimodal traffic simulation, route recommendation and subsidy: Enhancing first and last mile connectivity for MTR*", Jan 1 2026 – 31 Dec 2028, Core Member

HONORS AND AWARDS

2024 - 2029	Postgraduate Scholarship
09/2019	First Prize, Contemporary Undergraduate Mathematical Contest in Modeling (Beijing)
03/2019	Third Prize, BIT Science Contest
2017 & 2018	Second-Class Scholarship for Academic Excellence, Beijing Institute of Technology Postgraduate Scholarship